

the feedback loop

**Business
objective**

what are we
making better,
and for whom?

**ML
objective**

what is pre-
dicted, and what
may it get wrong?

**Proxy and
guardrail**

offline, on-
line, and the
gap between them

**Data and
labels**

where labels
come from,
how late, how leaky

**Feedback
loop**

drift, feed-
back effects,
harm, rollback

**Unit
cost**

per thousand
requests, per
trained model,
per token

**Serving
budget**

percentile,
batch, hardware,
and the arithmetic

**Model and
stages**

why this
many stages,
and why these